

Diagnostic information - DME (DFI) control unit

Oxygen sensor downstream of catalytic converter, bank 2

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

i INFORMATION

♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Perform all short tests in sequence

or

- Engine temperature > 60 °C (140 °F)
- Time since engine start > 10 second(s)
- Catalytic converter temperature 540 ... 750 °C (1,004 ... 1,382 °F)
- Drive for 60 seconds with slight acceleration at 40 ... 130 km/h (24 ... 80 mph)
- Active carbon filter loading < 0.7
- Engine speed < 3,000 rpm
- Lambda control ahead of catalytic converter active
- Lambda control behind catalytic converter active
- None of the following faults stored: Oxygen sensor(s) - electrical fault, intake manifold pressure sensor or throttle valve adjusting unit

Possible fault causes

♦ Oxygen sensor downstream of catalytic converter, bank 2 faulty

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♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Perform all short tests in sequence

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- Lambda control ahead of catalytic converter active
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- None of the following faults stored: Oxygen sensor(s) - electrical fault, intake manifold pressure sensor or throttle valve adjusting unit

Possible fault causes

♦ Oxygen sensor downstream of catalytic converter, bank 2 faulty

P2100

Diagnostic information - DME (DFI) control unit

Throttle valve adjusting unit - position deviation/control range

Diagnostic conditions

- Throttle valve adjusting unit adapted
- Ignition on

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit between lines (servo motor)
- ♦ Throttle valve adjusting unit faulty

Fault effects

The vehicle is in limp-home mode, i.e. the engine is running at 1,200 to 1,800 rpm.

P2103

Diagnostic information - DME (DFI) control unit

Throttle valve adjusting unit - position deviation/control range

Diagnostic conditions

- Throttle valve adjusting unit adapted
- Ignition on

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit between lines (servo motor)
- ♦ Throttle valve adjusting unit faulty

Fault effects

The vehicle is in limp-home mode, i.e. the engine is running at 1,200 to 1,800 rpm.

P2108

Diagnostic information - DME (DFI) control unit

Throttle valve adjusting unit

Diagnostic conditions

- Vehicle standstill
- Coolant temperature 5 ... 100 °C (41 ... 212 °F)
- Intake air temperature 5 ... 30 °C (41 ... 86 °F)
- Accelerator pedal not pressed (0%)
- Ignition off
- then switch on ignition > 60 second(s) with engine off

Possible fault causes

- ♦ Adaptation was not performed (successfully).
- ♦ Throttle valve adjusting unit faulty

P2119

Diagnostic information - DME (DFI) control unit

Throttle valve adjusting unit - start test

Diagnostic conditions

- **Throttle valve adjusting unit** adapted
- Switch on ignition , then
 - Switch off ignition for > 1 **min.**
 - Switch on ignition

Possible fault causes

- ♦ Throttle valve adjusting unit faulty

Fault setting condition(s)

The start test was faulty twice in a row (See ⓘ).

ⓘ **INFORMATION**

Start test: When the ignition is switched on , the throttle valve is opened by a certain value, and must then close on its own.

P2121

Diagnostic information - DME (DFI) control unit

Throttle valve potentiometer 1 implausible

Diagnostic conditions

- Engine running
- None of the following faults stored: P2127 or P2128 (throttle valve potentiometer 2)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines (potentiometer)
- ♦ Throttle valve adjusting unit faulty

Fault setting condition(s)

- Throttle valve potentiometer 1 (measured) - mass air flow (calculated)
⇒ Deviation too high

Fault effects

The vehicle is in limp-home mode, i.e. the engine is running at 1,200 to 1,800 rpm.

P2122

Diagnostic information - DME (DFI) control unit

Throttle potentiometer 1

Diagnostic conditions

- Ignition on > 5 second(s)
- then engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines (**potentiometer**)
- ♦ **Throttle valve adjusting unit** faulty

Fault effects

- ♦ No acceleration initially + increased idle speed (approx. 800 rpm)
 - ▶ until the accelerator pedal is released fully.
- ♦ Then acceleration, with
 - ♦ significantly reduced engine torque (max. approx. 100 ... 180 Nm)
 - ♦ and limited engine speed (max. approx. 5,000 rpm).
- ♦ If the accelerator and brake are pressed at the same time, the engine will run at idle speed.

P2123

Diagnostic information - DME (DFI) control unit

Throttle potentiometer 1

Diagnostic conditions

- Ignition on > 5 second(s)
- then engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines (**potentiometer**)
- ♦ **Throttle valve adjusting unit** faulty

Fault effects

- ♦ No acceleration initially + increased idle speed (approx. 800 rpm)
 - ▶ until the accelerator pedal is released fully.
- ♦ Then acceleration, with
 - ♦ significantly reduced engine torque (max. approx. 100 ... 180 Nm)
 - ♦ and limited engine speed (max. approx. 5,000 rpm).
- ♦ If the accelerator and brake are pressed at the same time, the engine will run at idle speed.

P2126

Diagnostic information - DME (DFI) control unit

Throttle valve potentiometer 2 implausible

Diagnostic conditions

- Engine running
- None of the following faults stored: P2122 or P2123 (throttle valve potentiometer 1)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines (potentiometer)
- ♦ Throttle valve adjusting unit faulty

Fault setting condition(s)

- Throttle valve potentiometer 2 (measured) - mass air flow (calculated)
⇒ Deviation too high

Fault effects

The vehicle is in limp-home mode, i.e. the engine is running at 1,200 to 1,800 rpm.

P2127

Diagnostic information - DME (DFI) control unit

Throttle potentiometer 2

Diagnostic conditions

- Ignition on > 5 second(s)
- then engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines (**potentiometer**)
- ♦ **Throttle valve adjusting unit** faulty

Fault effects

- ♦ No acceleration initially + increased idle speed (approx. 800 rpm)
 - ▶ until the accelerator pedal is released fully.
- ♦ Then acceleration, with
 - ♦ significantly reduced engine torque (max. approx. 100 ... 180 Nm)
 - ♦ and limited engine speed (max. approx. 5,000 rpm).
- ♦ If the accelerator and brake are pressed at the same time, the engine will run at idle speed.

P2128

Diagnostic information - DME (DFI) control unit

Throttle potentiometer 2

Diagnostic conditions

- Ignition on > 5 second(s)
- then engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines (**potentiometer**)
- ♦ **Throttle valve adjusting unit** faulty

Fault effects

- ♦ No acceleration initially + increased idle speed (approx. 800 rpm)
 - ▶ until the accelerator pedal is released fully.
- ♦ Then acceleration, with
 - ♦ significantly reduced engine torque (max. approx. 100 ... 180 Nm)
 - ♦ and limited engine speed (max. approx. 5,000 rpm).
- ♦ If the accelerator and brake are pressed at the same time, the engine will run at idle speed.

Diagnostic information - DME (DFI) control unit

Throttle valve adjusting unit - channel comparison

Diagnostic conditions

- Throttle valve adjusting unit adapted
- Idling and short acceleration
- then driving with different operating conditions

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Throttle valve adjusting unit faulty

Fault setting condition(s)

- The two potentiometer values differ too greatly from each other.
- It is not clear which value is incorrect.

Fault effects

- ♦ No throttle response
- ♦ Increased idle speed (approx. 1,000 1,600 rpm)
- ♦ Electronic throttle emergency mode (see ⓘ)

ⓘ INFORMATION

Electronic throttle emergency mode

During electronic throttle emergency mode, no power is supplied to the throttle valve, which then drops to emergency air position.

- ♦ Idle speed increases to approx. 1,000 1,200 rpm.
- ♦ If the accelerator pedal is pressed, the engine will run at max. 1,600 rpm (set via ignition, injection, ...). ► **Limp home**.
- ♦ If the accelerator and brake are pressed at the same time, the engine will run at idle speed.
- ♦ If the engine speed exceeds 1,600 rpm, cylinders will be deactivated.
- ♦ If cylinder deactivation is not sufficient, the DME control unit performs a reset and fault code **P1559** is stored in the fault memory (the original faults were previously stored as reset-resistant faults in the fault memory).

Diagnostic information - DME (DFI) control unit

Throttle valve adjusting unit - adaptation aborted due to environmental condition

Diagnostic conditions

- Adapt throttle valve using PIWIS Tester
- Then set conditions for the vehicle that prevent automatic adaptation of the throttle valve, i.e.:
 - Coolant temperature < 5 °C (41 °F) or > 100 °C (212 °F), or
 - Intake air temperature < 5 °C (41 °F) or > 30 °C (86 °F)
- Then wait for control unit run-on (fan run-on can take up to 25 minutes, depending on temperatures)
- Afterwards, the conditions specified above must still be set for the vehicle.
- Then start the engine

Possible fault causes

- ♦ Previous repairs were carried out during which:
 - ♦ the throttle housing was replaced, or
 - ♦ the DME control unit was programmed or replaced
- ♦ The control unit cannot enter run-on mode because the power supply is interrupted while the engine is running or immediately after switching off the ignition.
- ♦ The memory in the control unit is faulty

Fault setting condition(s)

No throttle valve adaptation values are stored in the DME control unit's permanent memory (EEPROM).

► See also 

Fault effects

- ♦ Engine cannot be started

INFORMATION

If no throttle valve adaptation values are stored, an adaptation is normally performed when the ignition is switched on or when starting the engine.

- ♦ Exception: The adaptation conditions are not met (e.g. outside temperature is too low) ► see

Adaptation conditions

- ♦ In this case, forced adaptation can be performed using the PIWIS Tester.

Adaptation conditions

- Vehicle is stationary
- Coolant temperature 5 ... 100 °C (41 ... 212 °F)
- Intake air temperature 5 ... 30 °C (41 ... 86 °F)
- Accelerator pedal not pressed (0%)
- Ignition off
- Then switch on ignition > 60 second(s) with engine off

P2177

Diagnostic information - DME (DFI) control unit

Lambda controller or lambda control adaptation, bank 1: system too lean

Diagnostic conditions

- Short test

or

- Lambda control active
- Lambda control adapted
- Engine temperature > 60 °C (140 °F)
- Intake air temperature < 80 °C (176 °F)
- Engine running > 45 second(s), at:
 - Engine speed < 800 rpm
- Then drive with different operating conditions in the lower and upper load ranges with valve lift switchover from small lift to large lift ([A220](#)).

Possible fault causes

- ♦ Leak in exhaust system (engine ⇒ oxygen sensor upstream of catalytic converter)
- ♦ Exhaust system constricted/blocked
- ♦ Fuel injector faulty (mechanical fault)
- ♦ Timing set incorrectly
- ♦ Fault in areas: fuel high pressure, intake manifold pressure sensor, throttle valve adjusting unit, ambient pressure sensor, intake air temperature sensor, rpm sensor, valve lift control, camshaft adjustment

Fault setting condition(s)

- P0171 - 30% deviation of the lambda controller for more than 30 seconds
- P2187 - 3.5 mg/stroke deviation in lambda control adaptation in the range near idle speed
- P2177 - 30% deviation in lambda control adaptation in the lower partial load range

INFORMATION

Opposed adaptation values in connection with misfires indicate that the timing is out of adjustment.

P2178

Diagnostic information - DME (DFI) control unit

Lambda controller or lambda control adaptation, bank 1: system too rich

Diagnostic conditions

- Short test

or

- Lambda control active
- Lambda control adapted
- Engine temperature > 60 °C (140 °F)
- Intake air temperature < 80 °C (176 °F)
- Engine running > 45 second(s), at:
 - Engine speed < 800 rpm
- Then drive with different operating conditions in the lower and upper load ranges with valve lift switchover from small lift to large lift ([A220](#)).

Possible fault causes

- ♦ High fuel content in engine oil.
- ♦ Leak in exhaust system (engine ⇒ oxygen sensor upstream of catalytic converter)
- ♦ Fuel injector faulty (mechanical fault)
- ♦ Tank vent valve stuck open
- ♦ Timing set incorrectly
- ♦ Fault in areas: fuel high pressure, intake manifold pressure sensor, throttle valve adjusting unit, ambient pressure sensor, intake air temperature sensor, rpm sensor, valve lift control, camshaft adjustment

Fault setting condition(s)

- P0172 - 30% deviation of the lambda controller for more than 30 seconds
- P2188 - 3.5 mg/stroke deviation in lambda control adaptation in the range near idle speed
- P2178 - 30% deviation in lambda control adaptation in the lower partial load range

INFORMATION

Opposed adaptation values in connection with misfires indicate that the timing is out of adjustment.

P2179

Diagnostic information - DME (DFI) control unit

Lambda controller or lambda control adaptation, bank 2: system too lean

Diagnostic conditions

- Short test

or

- Lambda control active
- Lambda control adapted
- Engine temperature > 60 °C (140 °F)
- Intake air temperature < 80 °C (176 °F)
- Engine running > 45 second(s), at:
 - Engine speed < 800 rpm
- Then drive with different operating conditions in the lower and upper load ranges with valve lift switchover from small lift to large lift ([A220](#)).

Possible fault causes

- ♦ Leak in exhaust system (engine ⇒ oxygen sensor upstream of catalytic converter)
- ♦ Exhaust system constricted/blocked
- ♦ Fuel injector faulty (mechanical fault)
- ♦ Timing set incorrectly
- ♦ Fault in areas: fuel high pressure, intake manifold pressure sensor, throttle valve adjusting unit, ambient pressure sensor, intake air temperature sensor, rpm sensor, valve lift control, camshaft adjustment

Fault setting condition(s)

- P0174 - 30% deviation of the lambda controller for more than 30 seconds
- P2189 - 3.5 mg/stroke deviation in lambda control adaptation in the range near idle speed
- P2179 - 30% deviation in lambda control adaptation in the lower partial load range

INFORMATION

Opposed adaptation values in connection with misfires indicate that the timing is out of adjustment.

P2180

Diagnostic information - DME (DFI) control unit

Lambda controller or lambda control adaptation, bank 2: system too rich

Diagnostic conditions

- Short test

or

- Lambda control active
- Lambda control adapted
- Engine temperature > 60 °C (140 °F)
- Intake air temperature < 80 °C (176 °F)
- Engine running > 45 second(s), at:
 - Engine speed < 800 rpm
- Then drive with different operating conditions in the lower and upper load ranges with valve lift switchover from small lift to large lift ([A220](#)).

Possible fault causes

- ♦ High fuel content in engine oil.
- ♦ Leak in exhaust system (engine ⇒ oxygen sensor upstream of catalytic converter)
- ♦ Fuel injector faulty (mechanical fault)
- ♦ Tank vent valve stuck open
- ♦ Timing set incorrectly
- ♦ Fault in areas: fuel high pressure, intake manifold pressure sensor, throttle valve adjusting unit, ambient pressure sensor, intake air temperature sensor, rpm sensor, valve lift control, camshaft adjustment

Fault setting condition(s)

- P0175 - 30% deviation of the lambda controller for more than 30 seconds
- P2190 - 3.5 mg/stroke deviation in lambda control adaptation in the range near idle speed
- P2180 - 30% deviation in lambda control adaptation in the lower partial load range

INFORMATION

Opposed adaptation values in connection with misfires indicate that the timing is out of adjustment.

Diagnostic information - DME (DFI) control unit

Thermostat stuck open

Diagnostic conditions

- Perform Thermostat system test

or

- Driving under the following conditions until the model coolant temperature has reached 74 °C (165.2 °F):
 - **Engine start at**
 - Outside temperature (T050) > -7 °C (19.4 °F)
 - and
 - Engine coolant temperature (T020) < 55 °C (131 °F)
 - **Driving with:**
 - Vehicle speed of between 25 ... 150 km/h (15 ... 93 mph) for at least 35% of drive
 - Total share of overrun phases < 40%
 - Share of Start Stop engine stop phases < 50%

Possible fault causes

- ♦ Combination valve for cooling system bleeding not closed (service bow is up)
- ♦ Combination valve for cooling system bleeding leaking
- ♦ Thermostat faulty
- ♦ Seal in thermostat housing faulty
- ♦ Thermostat housing faulty

Fault setting condition(s)

- The measured engine coolant temperature is less than 74 °C (165.2 °F) when the temperature model has reached 74 °C (165.2 °F).

or

- Engine coolant temperature (T020) drops to below 65 °C (149 °F) for longer than 30 seconds.

Fault effects

- ♦ The radiator shutters (if fitted) are opened and remain open during the next drives until the fault has been healed or deleted from the fault memory.

P2184

Diagnostic information - DME (DFI) control unit

Radiator outlet coolant temperature sensor: electrical fault

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Radiator outlet coolant temperature sensor faulty

Fault setting condition(s)

- Coolant temperature at radiator outlet (measured) < -40 °C (-40 °F) (P2184), with :
 - Ignition on or engine running
 - and
 - Intake air temperature > -30 °C (-22 °F)
- or
- Engine running > 31 minute(s)
- or
- Coolant temperature at radiator outlet (measured) > +140 °C (284 °F) (P2185), with :
 - Ignition on or engine running

P2185

Diagnostic information - DME (DFI) control unit

Radiator outlet coolant temperature sensor: electrical fault

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Radiator outlet coolant temperature sensor faulty

Fault setting condition(s)

- Coolant temperature at radiator outlet (measured) < -40 °C (-40 °F) (P2184), with :
 - Ignition on or engine running
 - and
 - Intake air temperature > -30 °C (-22 °F)
- or
- Engine running > 31 minute(s)
- or
- Coolant temperature at radiator outlet (measured) > +140 °C (284 °F) (P2185), with :
 - Ignition on or engine running

P2186

Diagnostic information - DME (DFI) control unit

Radiator outlet coolant temperature sensor: loose contact

Diagnostic conditions

- Ignition on
- None of the following faults stored: P2184, P2185

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Radiator outlet coolant temperature sensor faulty

Fault setting condition(s)

Coolant temperature at radiator outlet: actual-value jumps > 3 °C (37 °F)

Fault effects

- ♦ Substitute value = last actual value that was detected as OK, with:
 - Actual-value jumps > 9.75 °C (49.5 °F)

P2187

Diagnostic information - DME (DFI) control unit

Lambda controller or lambda control adaptation, bank 1: system too lean

Diagnostic conditions

- Short test

or

- Lambda control active
- Lambda control adapted
- Engine temperature > 60 °C (140 °F)
- Intake air temperature < 80 °C (176 °F)
- Engine running > 45 second(s), at:
 - Engine speed < 800 rpm
- Then drive with different operating conditions in the lower and upper load ranges with valve lift switchover from small lift to large lift ([A220](#)).

Possible fault causes

- ♦ Leak in exhaust system (engine ⇒ oxygen sensor upstream of catalytic converter)
- ♦ Exhaust system constricted/blocked
- ♦ Fuel injector faulty (mechanical fault)
- ♦ Timing set incorrectly
- ♦ Fault in areas: fuel high pressure, intake manifold pressure sensor, throttle valve adjusting unit, ambient pressure sensor, intake air temperature sensor, rpm sensor, valve lift control, camshaft adjustment

Fault setting condition(s)

- P0171 - 30% deviation of the lambda controller for more than 30 seconds
- P2187 - 3.5 mg/stroke deviation in lambda control adaptation in the range near idle speed
- P2177 - 30% deviation in lambda control adaptation in the lower partial load range

INFORMATION

Opposed adaptation values in connection with misfires indicate that the timing is out of adjustment.

P2188

Diagnostic information - DME (DFI) control unit

Lambda controller or lambda control adaptation, bank 1: system too rich

Diagnostic conditions

- Short test

or

- Lambda control active
- Lambda control adapted
- Engine temperature > 60 °C (140 °F)
- Intake air temperature < 80 °C (176 °F)
- Engine running > 45 second(s), at:
 - Engine speed < 800 rpm
- Then drive with different operating conditions in the lower and upper load ranges with valve lift switchover from small lift to large lift ([A220](#)).

Possible fault causes

- ♦ High fuel content in engine oil.
- ♦ Leak in exhaust system (engine ⇒ oxygen sensor upstream of catalytic converter)
- ♦ Fuel injector faulty (mechanical fault)
- ♦ Tank vent valve stuck open
- ♦ Timing set incorrectly
- ♦ Fault in areas: fuel high pressure, intake manifold pressure sensor, throttle valve adjusting unit, ambient pressure sensor, intake air temperature sensor, rpm sensor, valve lift control, camshaft adjustment

Fault setting condition(s)

- P0172 - 30% deviation of the lambda controller for more than 30 seconds
- P2188 - 3.5 mg/stroke deviation in lambda control adaptation in the range near idle speed
- P2178 - 30% deviation in lambda control adaptation in the lower partial load range

INFORMATION

Opposed adaptation values in connection with misfires indicate that the timing is out of adjustment.

Diagnostic information - DME (DFI) control unit

Lambda controller or lambda control adaptation, bank 2: system too lean

Diagnostic conditions

- Short test

or

- Lambda control active
- Lambda control adapted
- Engine temperature > 60 °C (140 °F)
- Intake air temperature < 80 °C (176 °F)
- Engine running > 45 second(s), at:
 - Engine speed < 800 rpm
- Then drive with different operating conditions in the lower and upper load ranges with valve lift switchover from small lift to large lift ([A220](#)).

Possible fault causes

- ♦ Leak in exhaust system (engine ⇒ oxygen sensor upstream of catalytic converter)
- ♦ Exhaust system constricted/blocked
- ♦ Fuel injector faulty (mechanical fault)
- ♦ Timing set incorrectly
- ♦ Fault in areas: fuel high pressure, intake manifold pressure sensor, throttle valve adjusting unit, ambient pressure sensor, intake air temperature sensor, rpm sensor, valve lift control, camshaft adjustment

Fault setting condition(s)

- P0174 - 30% deviation of the lambda controller for more than 30 seconds
- P2189 - 3.5 mg/stroke deviation in lambda control adaptation in the range near idle speed
- P2179 - 30% deviation in lambda control adaptation in the lower partial load range

INFORMATION

Opposed adaptation values in connection with misfires indicate that the timing is out of adjustment.

P2190

Diagnostic information - DME (DFI) control unit

Lambda controller or lambda control adaptation, bank 2: system too rich

Diagnostic conditions

- Short test

or

- Lambda control active
- Lambda control adapted
- Engine temperature > 60 °C (140 °F)
- Intake air temperature < 80 °C (176 °F)
- Engine running > 45 second(s), at:
 - Engine speed < 800 rpm
- Then drive with different operating conditions in the lower and upper load ranges with valve lift switchover from small lift to large lift ([A220](#)).

Possible fault causes

- ♦ High fuel content in engine oil.
- ♦ Leak in exhaust system (engine ⇒ oxygen sensor upstream of catalytic converter)
- ♦ Fuel injector faulty (mechanical fault)
- ♦ Tank vent valve stuck open
- ♦ Timing set incorrectly
- ♦ Fault in areas: fuel high pressure, intake manifold pressure sensor, throttle valve adjusting unit, ambient pressure sensor, intake air temperature sensor, rpm sensor, valve lift control, camshaft adjustment

Fault setting condition(s)

- P0175 - 30% deviation of the lambda controller for more than 30 seconds
- P2190 - 3.5 mg/stroke deviation in lambda control adaptation in the range near idle speed
- P2180 - 30% deviation in lambda control adaptation in the lower partial load range

INFORMATION

Opposed adaptation values in connection with misfires indicate that the timing is out of adjustment.

Diagnostic information - DME (DFI) control unit

Oxygen sensor upstream of catalytic converter, bank 1 - signal too lean/too rich

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

INFORMATION

- ♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.
- ♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Perform all short tests in sequence (without tank leak test).

or

- Engine running > 3 minute(s) with:
 - No overrun phase(s)
 - All oxygen sensor(s) ready for operation
 - Lambda control active
 - Engine speed 640 ... 4,000 rpm
 - Load 15 ... 70%
 - Catalytic converter temperature > 400 °C (752 °F)
 - Mass air flow 10 ... 520 kg/h
- None of the following faults stored: P2270, P2271, P0041

Possible fault causes

- ♦ Oxygen sensor upstream of catalytic converter bank 1 faulty

Fault setting condition(s)

- Signal deviation as a result of broken ceramic insulator

P2196

Diagnostic information - DME (DFI) control unit

Oxygen sensor upstream of catalytic converter, bank 1 - signal too lean/too rich

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

INFORMATION

- ♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.
- ♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Perform all short tests in sequence (without tank leak test).

or

- Engine running > 3 minute(s) with:
 - No overrun phase(s)
 - All oxygen sensor(s) ready for operation
 - Lambda control active
 - Engine speed 640 ... 4,000 rpm
 - Load 15 ... 70%
 - Catalytic converter temperature > 400 °C (752 °F)
 - Mass air flow 10 ... 520 kg/h
- None of the following faults stored: P2270, P2271, P0041

Possible fault causes

- ♦ Oxygen sensor upstream of catalytic converter bank 1 faulty

Fault setting condition(s)

- Signal deviation as a result of broken ceramic insulator

Diagnostic information - DME (DFI) control unit

Oxygen sensor upstream of catalytic converter, bank 2 - signal too lean/too rich

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

INFORMATION

- ♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.
- ♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Perform all short tests in sequence (without tank leak test).

or

- Engine running > 3 minute(s) with:
 - No overrun phase(s)
 - All oxygen sensor(s) ready for operation
 - Lambda control active
 - Engine speed 640 ... 4,000 rpm
 - Load 15 ... 70%
 - Catalytic converter temperature > 400 °C (752 °F)
 - Mass air flow 10 ... 520 kg/h
- None of the following faults stored: P2270, P2271, P0041

Possible fault causes

- ♦ Oxygen sensor upstream of catalytic converter bank 2 faulty

Fault setting condition(s)

- Signal deviation as a result of broken ceramic insulator

Diagnostic information - DME (DFI) control unit

Oxygen sensor upstream of catalytic converter, bank 2 - signal too lean/too rich

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

INFORMATION

- ♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.
- ♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Perform all short tests in sequence (without tank leak test).

or

- Engine running > 3 minute(s) with:
 - No overrun phase(s)
 - All oxygen sensor(s) ready for operation
 - Lambda control active
 - Engine speed 640 ... 4,000 rpm
 - Load 15 ... 70%
 - Catalytic converter temperature > 400 °C (752 °F)
 - Mass air flow 10 ... 520 kg/h
- None of the following faults stored: P2270, P2271, P0041

Possible fault causes

- ♦ Oxygen sensor upstream of catalytic converter bank 2 faulty

Fault setting condition(s)

- Signal deviation as a result of broken ceramic insulator

P2227

Diagnostic information - DME (DFI) control unit

Ambient pressure sensor

Diagnostic conditions

- Ignition on > 10 second(s)
- Then: Acceleration with wide-open throttle for as long as possible, immediately followed by an overrun phase lasting as long as possible

Possible fault causes

- ♦ DME (DFI) control unit faulty

Fault effects

- ♦ Substitute value from model

P2228

Diagnostic information - DME (DFI) control unit

Ambient pressure sensor

Diagnostic conditions

- Ignition on > 10 second(s)
- Then: Acceleration with wide-open throttle for as long as possible, immediately followed by an overrun phase lasting as long as possible

Possible fault causes

- ♦ DME (DFI) control unit faulty

Fault effects

- ♦ Substitute value from model

P2229

Diagnostic information - DME (DFI) control unit

Ambient pressure sensor

Diagnostic conditions

- Ignition on > 10 second(s)
- Then: Acceleration with wide-open throttle for as long as possible, immediately followed by an overrun phase lasting as long as possible

Possible fault causes

- ♦ DME (DFI) control unit faulty

Fault effects

- ♦ Substitute value from model

P2230

Diagnostic information - DME (DFI) control unit

Ambient pressure sensor

Diagnostic conditions

- Ignition on > 10 second(s)
- Then: Acceleration with wide-open throttle for as long as possible, immediately followed by an overrun phase lasting as long as possible

Possible fault causes

- ♦ DME (DFI) control unit faulty

Fault effects

- ♦ Substitute value from model

Diagnostic information - DME (DFI) control unit

Oxygen sensor upstream of catalytic converter, bank 1

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

INFORMATION

♦ Do not interchange the oxygen sensor upstream of the catalytic converter with the sensor downstream of the catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Idling > 3 second(s)

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line
- ♦ Oxygen sensor upstream of catalytic converter bank 1 faulty

Diagnostic information - DME (DFI) control unit

Oxygen sensor upstream of catalytic converter, bank 2

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

i INFORMATION

♦ Do not interchange the oxygen sensor upstream of the catalytic converter with the sensor downstream of the catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Idling > 3 second(s)

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line
- ♦ Oxygen sensor upstream of catalytic converter bank 2 faulty

Diagnostic information - DME (DFI) control unit

Oxygen sensor upstream of catalytic converter, bank 1

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

i INFORMATION

♦ Do not interchange the oxygen sensor upstream of the catalytic converter with the sensor downstream of the catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Idling > 3 second(s)

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line
- ♦ Oxygen sensor upstream of catalytic converter bank 1 faulty

Diagnostic information - DME (DFI) control unit

Oxygen sensor upstream of catalytic converter, bank 2

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

INFORMATION

♦ Do not interchange the oxygen sensor upstream of the catalytic converter with the sensor downstream of the catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Idling > 3 second(s)

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line
- ♦ Oxygen sensor upstream of catalytic converter bank 2 faulty

Diagnostic information - DME (DFI) control unit

Oxygen sensor upstream of catalytic converter, bank 1

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

i INFORMATION

♦ Do not interchange the oxygen sensor upstream of the catalytic converter with the sensor downstream of the catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Idling > 3 second(s)

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line
- ♦ Oxygen sensor upstream of catalytic converter bank 1 faulty

Diagnostic information - DME (DFI) control unit

Oxygen sensor upstream of catalytic converter, bank 2

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

i INFORMATION

♦ Do not interchange the oxygen sensor upstream of the catalytic converter with the sensor downstream of the catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Idling > 3 second(s)

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line
- ♦ Oxygen sensor upstream of catalytic converter bank 2 faulty